



A BLUE TEAM STORYBOOK

OSI MODEL

EXPLAINED LIKE A CYBER CRIME MOVIE



A Story-Based Guide for
SOC ANALYSTS & BLUE TEAMERS

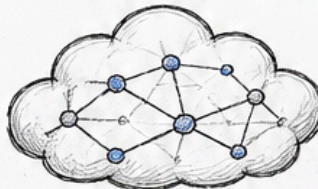
THE HACKER
(THE VILLAIN)



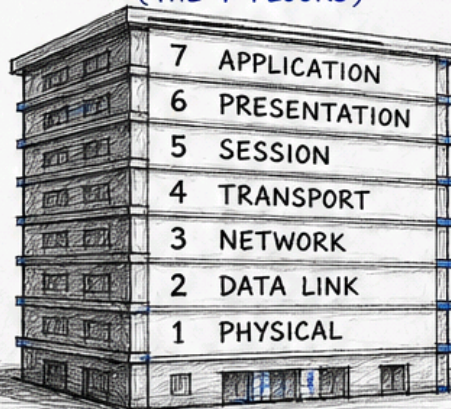
THE BLUE TEAM ANALYST
(THE HERO)



THE INTERNET
(THE HIGHWAY)



THE 7 LAYERS
(THE 7 FLOORS)



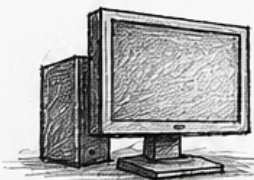
THE MISSION

A secret message
is sent across
the network.
The hacker wants
to steal it.
The Blue Team
must protect it.

OUR GOAL

- ✓ Understand each layer
- ✓ See how data travels
- ✓ Detect attacks
- ✓ Protect systems
- ✓ Become better defenders

SENDER
(THE SOURCE)



RECEIVER
(THE DESTINATION)



EVERY MESSAGE TRAVELS THROUGH SEVEN LAYERS
BEFORE REACHING ITS DESTINATION.



KNOW THE JOURNEY.
Understand how data
moves across networks.



SEE THE ATTACKS.
Learn where attackers
intercept and abuse.



INVESTIGATE BETTER.
Use OSI knowledge to
analyze and detect.



DEFEND SMARTER.
Stronger defenses start
with deeper understanding.



BLUETEAMERS
DEFEND. DETECT. RESPOND.



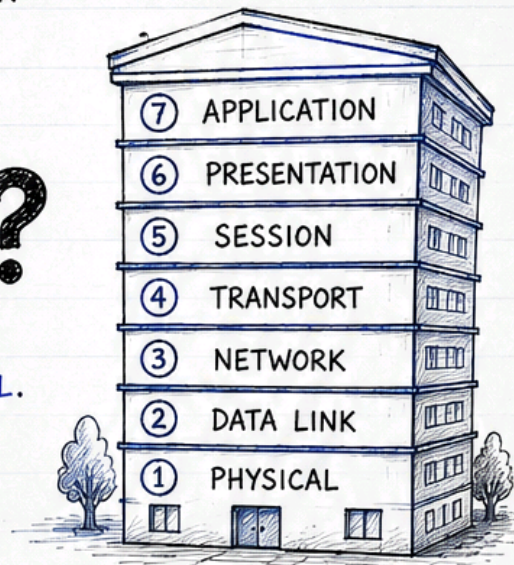
blueteamers.io



WHAT IS OSI MODEL?

OSI stands for **OPEN SYSTEMS INTERCONNECTION MODEL**.

It is a conceptual framework that describes how data travels from one device to another across a network using **seven** different layers.



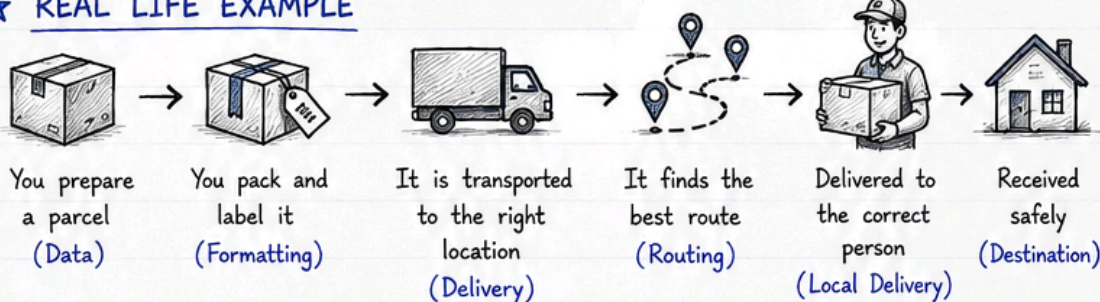
★ WHY DO WE USE OSI MODEL?

- ✓ Provides a standard way for devices to communicate.
- ✓ Makes networking easier to understand.
- ✓ Helps in troubleshooting network problems.
- ✓ Improves network design and development.
- ✓ Helps in analyzing and securing network traffic.



Think of OSI as a 7-floor building. Each floor has a specific job!

★ REAL LIFE EXAMPLE



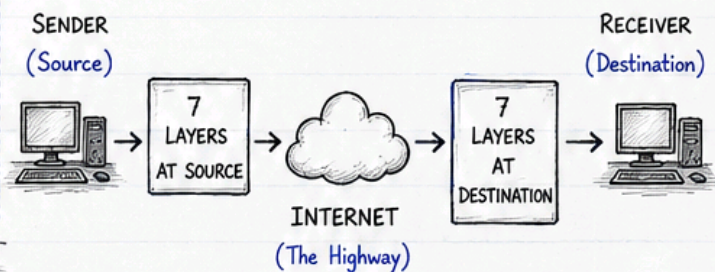
OSI works exactly like this process!

★ WHY SOC ANALYSTS LEARN OSI?

- Investigate network attacks
- Understand packet flow and behavior
- Analyze logs and alerts better
- Detect malicious activity at different layers
- Troubleshoot network issues faster



★ SIMPLE FLOW OF COMMUNICATION



KEY TAKEAWAY

OSI Model is the **roadmap** that shows how data travels from one computer to another. If you understand OSI, you understand networking!

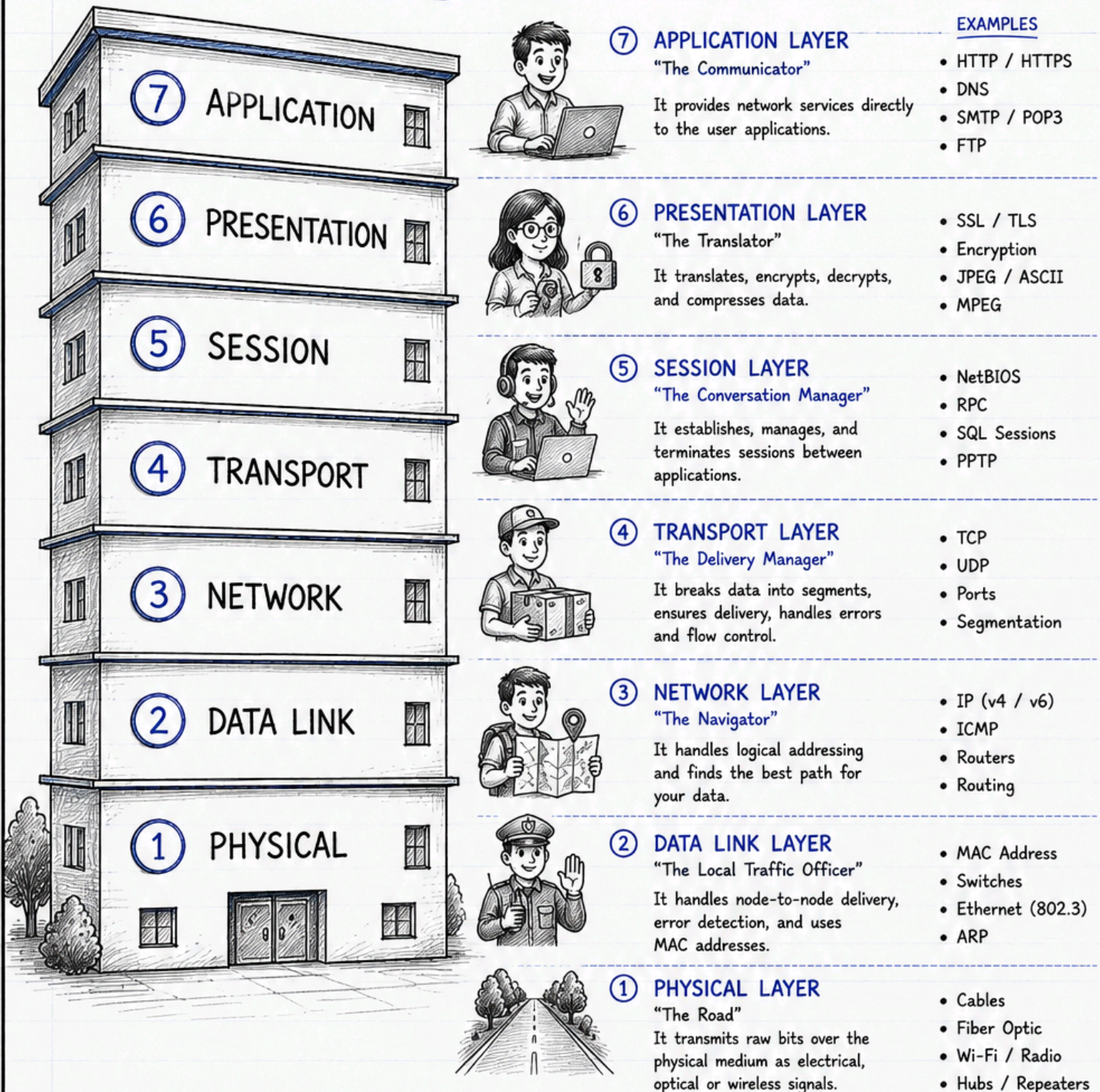




MEET THE 7 LAYERS

Think of OSI Model as a 7-floor building.

Each floor (layer) has a specific role to play in delivering your message safely.



★ KEY TAKEAWAY

Each layer **depends** on the one below it and **supports** the one above it. Together, they ensure that your data reaches the destination safely.





MEET THE CHARACTERS



Every movie needs heroes, villains and a mission.
Here is our story.

ALEX (THE HERO)



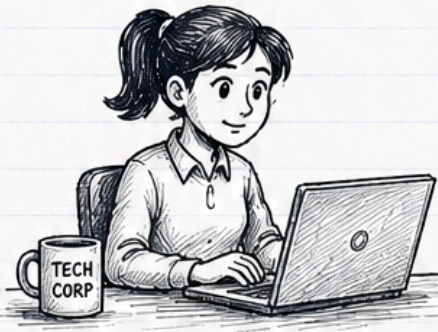
- Blue Team Analyst
- Protects systems
- Investigates attacks
- Uses knowledge to defend

DARKGHOST (THE VILLAIN)



- Cyber Criminal
- Wants to steal data
- Exploits weaknesses
- Works in the shadows

TECHCORP EMPLOYEE (THE VICTIM)



- Sends a confidential message
- Trusts the network
- Unaware of the danger
- Needs protection

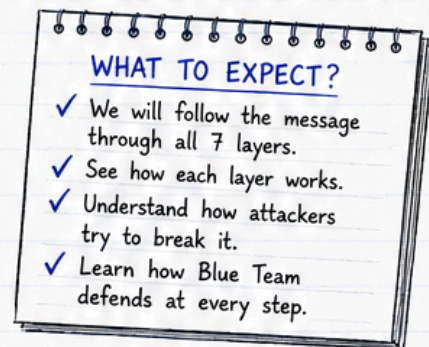


THE MISSION

A secret and important message must travel from the TechCorp employee to the bank server.
DarkGhost wants to steal it.
Alex must protect it and make sure it reaches safely!



- ★ The TechCorp employee writes a message.
- ★ The message starts its journey across the network.
- ★ DarkGhost is waiting to intercept it.
- ★ Alex will follow the message through every layer and stop the attack.



THE STORY JOURNEY



KEY TAKEAWAY

Before a message reaches its destination, it must go through 7 different layers.
Let's follow the journey and uncover every step!





SCENE 1: APPLICATION LAYER

"THE USER CREATES THE MESSAGE"



★ WHAT IS THE APPLICATION LAYER?

The Application Layer is the top layer (Layer 7) of the OSI Model.

It is the layer where **users interact** with network applications.

It provides **network services** directly to applications that people use every day.

IN THE MOVIE...

TechCorp employee Neha needs to send a confidential transfer request to the bank.



She opens her email application and writes the message.

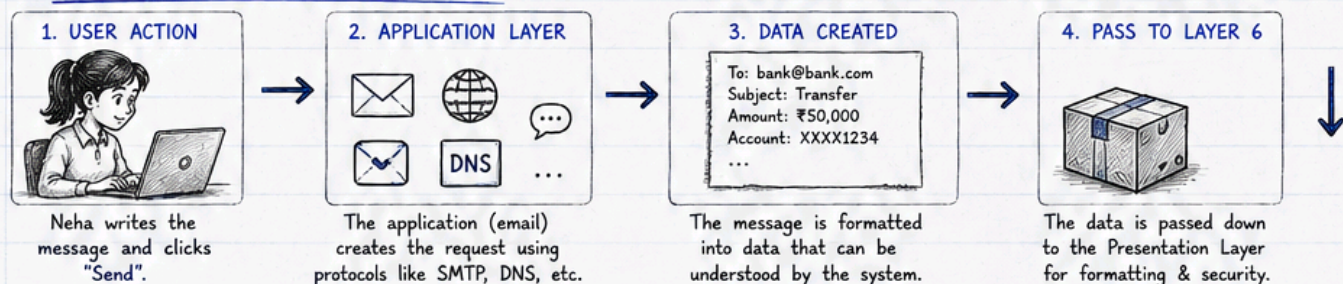
★ MAIN RESPONSIBILITIES

- Provides an interface for users to access network services.
- Handles user requests (web browsing, email, file transfer, etc.).
- Identifies the type of communication partners.
- Manages application level protocols.
- Ensures the data is ready for the lower layers.

★ COMMON APPLICATION LAYER PROTOCOLS

	HTTP / HTTPS	For web browsing (websites).
	SMTP / POP3 / IMAP	For sending and receiving emails.
	DNS	For resolving domain names (example.com → IP address).
	FTP / FTPS	For transferring files.
	SNMP	For network monitoring and management.
	SIP / RTP	For Voice over IP and video communication.

★ HOW IT WORKS (IN SIMPLE WORDS)



★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts monitor:

- ✓ Web traffic (HTTP/HTTPS)
- ✓ Email activity (SMTP/POP3)
- ✓ DNS queries and lookups
- ✓ Suspicious application usage
- ✓ Unusual login attempts

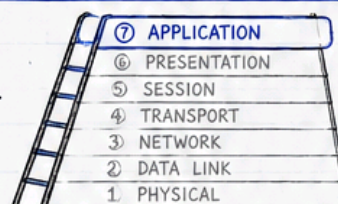
IN THE MOVIE...

DarkGhost is also watching... He sees Neha is sending **something**. But he can't read the content yet, because it's still in the upper layers!



KEY TAKEAWAY

The Application Layer is where everything begins. It takes the user's request, creates the data, and passes it down the ladder of layers.



The journey starts here!





SCENE 2:

PRESENTATION LAYER

"THE TRANSLATOR PROTECTS THE MESSAGE"



★ WHAT IS THE PRESENTATION LAYER?

The Presentation Layer is **Layer 6** of the OSI Model.

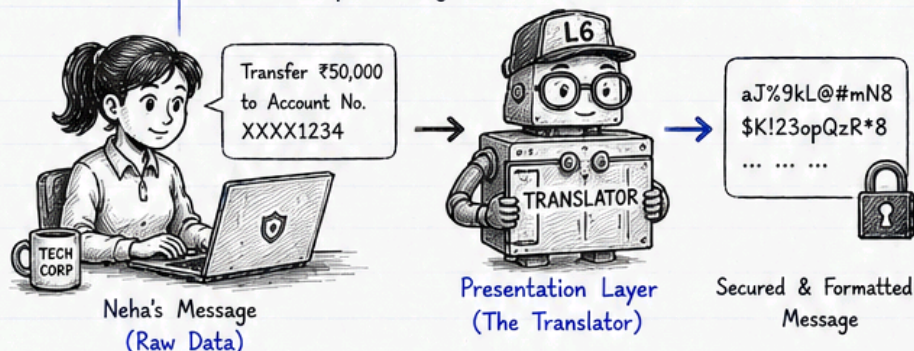
It acts as a **translator** between the Application Layer and the Session Layer.

It makes sure the data is in a format that the receiving system can **understand**.

It also **protects** the data using **encryption** and can **compress** it to save bandwidth.

IN THE MOVIE...

After Neha writes her transfer request, the message needs to be secured and formatted before the conversation between computers begins.



★ MAIN RESPONSIBILITIES

- Translates data into a common format.
- Encrypts data to keep it safe.
- Decrypts data at the receiving end.
- Compresses data to reduce size.
- Ensures data is readable and usable by the next layer.



★ KEY FUNCTIONS (IN SIMPLE WORDS)

Aॐ	Translation	Converts data from one format to another (e.g., ASCII, UTF-8).
🔒	Encryption	Scrambles data so attackers cannot read it.
🔓	Decryption	Turns encrypted data back into readable form.
📄	Compression	Reduces the size of data to save bandwidth.
✓	Formatting	Ensures data structure is correct for the receiving system.

★ HOW IT WORKS (IN SIMPLE STEPS)



★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts look for:

- ✓ Unusual encryption
- ✓ Weak or outdated ciphers
- ✓ Data exfiltration attempts
- ✓ Abnormal data compression
- ✓ Encoding anomalies

IN THE MOVIE...

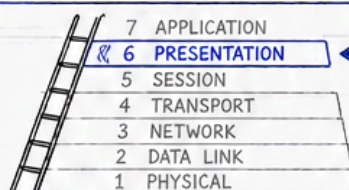
DarkGhost tries to intercept the message, but the Presentation Layer has already encrypted it.

He sees only gibberish and can't understand anything!



KEY TAKEAWAY

The Presentation Layer makes sure the data is in the right format, secure, and efficient before it moves forward.



It translates, protects and prepares the data for safe communication.





SCENE 3:

SESSION LAYER

"THE CONVERSATION BEGINS AND ENDS"



★ WHAT IS THE SESSION LAYER?

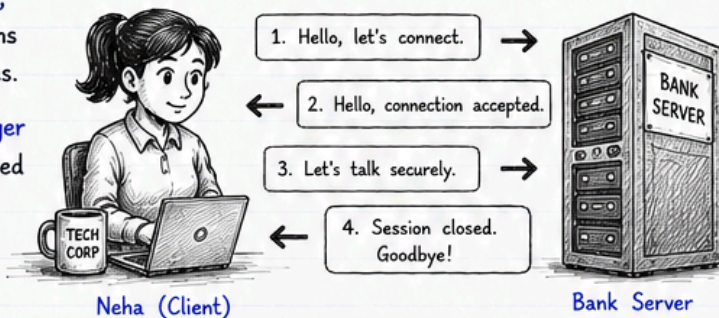
The Session Layer is **Layer 5** of the OSI Model.

It is responsible for **establishing**, **managing**, and **terminating** sessions between two communicating devices.

It acts like a **conversation manager** that keeps the connection organized from start to end.

IN THE MOVIE...

After the message is secured and formatted, Neha's computer needs to start a conversation with the bank server.



THE SESSION FLOW

- ✓ Start the session (handshake)
- ✓ Maintain the session (data exchange)
- ✓ Synchronize the conversation
- ✓ End the session (cleanly close)

★ MAIN RESPONSIBILITIES

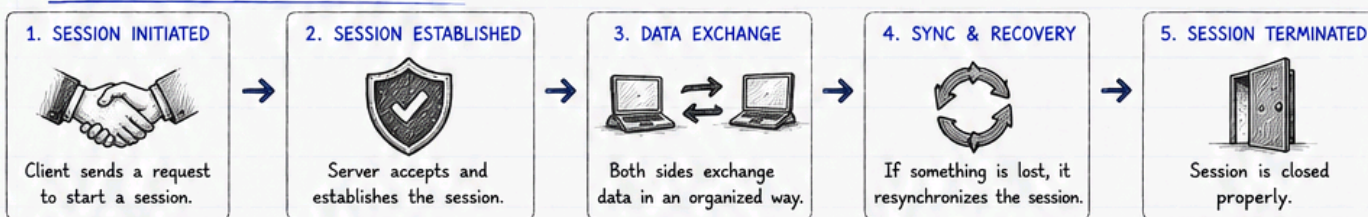
- Establishes a session between two devices.
- Manages the session by controlling dialogue (who speaks when).
- Synchronizes data if the connection is interrupted.
- Terminates the session properly when communication is complete.
- Supports full-duplex, half-duplex and simplex communication.



★ EXAMPLES OF SESSION LAYER PROTOCOLS

PROTOCOL	DESCRIPTION
NetBIOS	Manages sessions between computers in a network.
RPC	Remote Procedure Call - allows programs to execute on remote systems.
SQL Sessions	Manages database connections between client and server.
PPTP	Creates and manages VPN sessions.

★ HOW IT WORKS (IN SIMPLE STEPS)



★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts monitor:

- ✓ Unusual or long sessions
- ✓ Multiple failed sessions
- ✓ Abnormal resets
- ✓ Session hijacking attempts
- ✓ Excessive reconnections

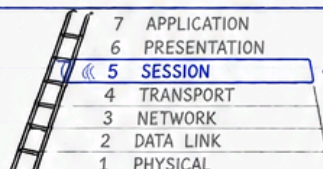
IN THE MOVIE...

DarkGhost tries to hijack the session to sneak into the conversation. But the Session Layer controls the flow and protects the dialogue between both sides!



KEY TAKEAWAY

The Session **Layer** starts, manages, synchronizes, and ends conversations between two systems. It keeps the communication organized and reliable.



It manages the conversation from start to finish.





SCENE 4:

TRANSPORT LAYER

"THE DELIVERY MANAGER SPLITS THE MESSAGE"



★ THE STORY SO FAR...

Neha's message has been created (Application Layer), protected (Presentation Layer) and the session is established (Session Layer).

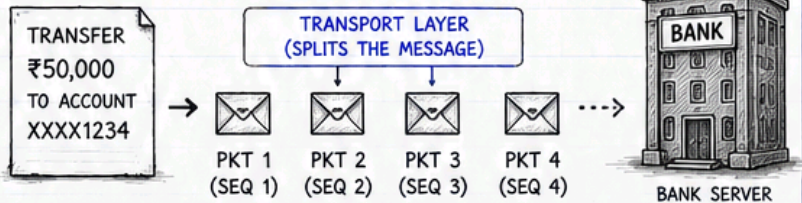
Now it's time to actually deliver the data reliably from one computer to another.

Meet the **Transport Layer** - the Delivery Manager!



IN THE MOVIE...

Neha's message is ready. But it's too big to send all at once. The Transport Layer breaks it into smaller pieces (segments). Each piece gets a number so they can be reassembled at the other end. If a piece is lost, it can be sent again.



★ WHAT IS TRANSPORT LAYER?

The Transport Layer is **Layer 4** of the OSI Model. It provides end-to-end communication services for applications.

It is responsible for:

- ✓ End-to-End Communication
- ✓ Segmentation and Reassembly
- ✓ Reliability and Error Handling
- ✓ Flow Control
- ✓ Port Numbering (Multiplexing)



I make sure every piece reaches safely and in order!

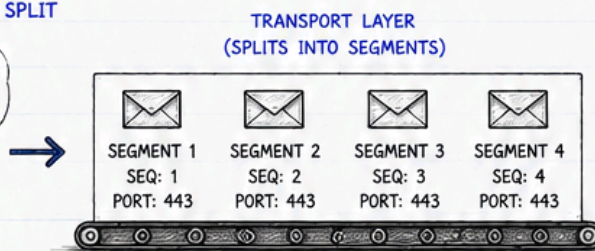
★ MAIN CONCEPTS

	TCP	Reliable, connection-oriented communication.
	UDP	Fast, connectionless communication.
	PORT NUMBERS	Identify applications on a device. (e.g., 80, 443, 53)
	SEGMENTATION	Breaks large data into smaller segments.
	REASSEMBLY	Combines segments back into original data.

★ HOW IT WORKS (IN SIMPLE STEPS)

1. Receives data from Session Layer.
2. Splits data into segments.
3. Adds sequence numbers and port numbers.
4. Chooses protocol (TCP / UDP) to deliver the segments.
5. Passes segments to Network Layer.

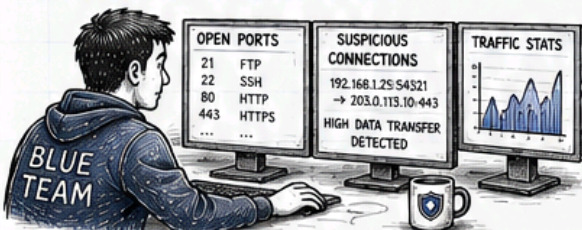
MOVIE SCENE: THE MESSAGE IS SPLIT



BANK SERVER (REASSEMBLES)



★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts monitor:

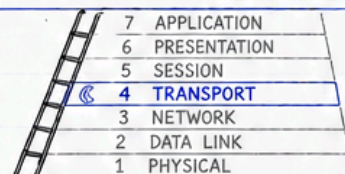
- ✓ Suspicious open ports
- ✓ Port scanning attempts
- ✓ Unusual connections
- ✓ High data transfers
- ✓ Abnormal protocols usage

IN THE MOVIE...



KEY TAKEAWAY

The Transport Layer is the Delivery Manager. It breaks the message into pieces, makes sure they are delivered correctly, and reassembles them at the destination.



You are here!
The message delivery starts here.





SCENE 5:

TCP vs UDP

"THE TWO DELIVERY COMPANIES"

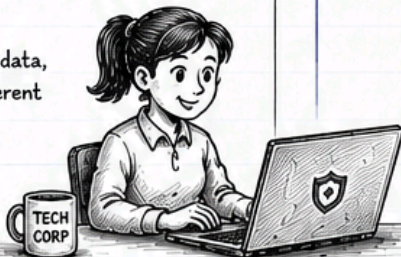


★ THE STORY SO FAR...

The Transport Layer is in charge of delivering the message from source to destination.

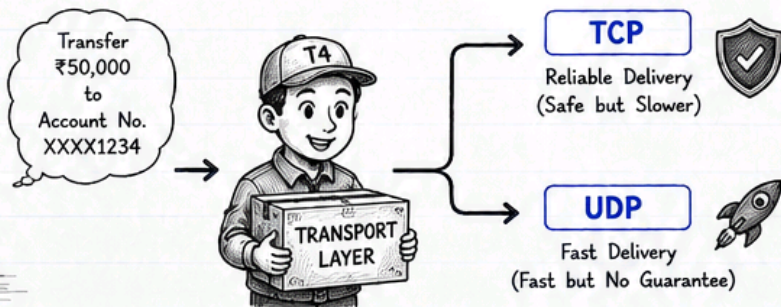
It has two delivery services – TCP and UDP.

Both help move data, but in very different ways!



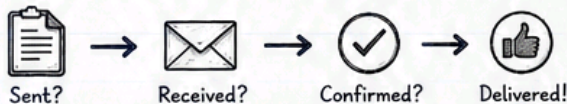
IN THE MOVIE...

Neha wants to send ₹50,000 to her bank. The Transport Layer chooses how the data will travel – using TCP or UDP.



TCP – THE RELIABLE COURIER

TCP ensures the data reaches safely. It checks, confirms and retransmits if needed.



KEY FEATURES

- Connection-oriented
- Reliable delivery
- Error checking & recovery
- Flow control
- Guarantees order of data

EXAMPLES

- HTTPS (Web Browsing)
- Email (SMTP/POP3)
- Banking Transactions
- File Transfers (FTP)



UDP – THE FAST COURIER

UDP just sends the data and moves on. No connection. No confirmation.



KEY FEATURES

- Connectionless
- Faster delivery
- No delivery guarantee
- No error recovery
- No order guarantee

EXAMPLES

- Online Gaming
- Voice Calls (VoIP)
- Live Streaming
- DNS Queries

IN THE MOVIE...



Neha is sending money to her bank. She chooses TCP because losing ₹50,000 is NOT an option!



HACKER SIDE...



DarkGhost prefers UDP for his attacks. It's faster, has less checking and is easier to abuse!



WHY HACKERS LOVE UDP?

- No connection setup
- No handshake
- No acknowledgement
- Harder to track
- Can be used for DDoS

★ TCP vs UDP – AT A GLANCE

FEATURE	TCP	UDP
Type	Connection-Oriented	Connectionless
Reliability	Reliable	Unreliable
Speed	Slower (Safe)	Faster
Error Recovery	Yes	No
Order Guarantee	Yes	No
Best For	Critical Data	Time-Sensitive Data



★ PORT NUMBERS – THE DELIVERY ADDRESSES

Applications are identified using Port Numbers.

80	443	25	53	...
HTTP	HTTPS	SMTP	DNS	



Port Numbers help the Transport Layer deliver the data to the correct application on the destination system.

KEY TAKEAWAY



TCP is like a careful courier – it makes sure the package arrives safely.
UDP is like a speedy courier – it delivers fast, but doesn't wait for confirmation.

“Choose TCP when accuracy matters.
Choose UDP when speed matters.”



SCENE 6: NETWORK LAYER

"THE NAVIGATOR FINDS THE ROUTE"



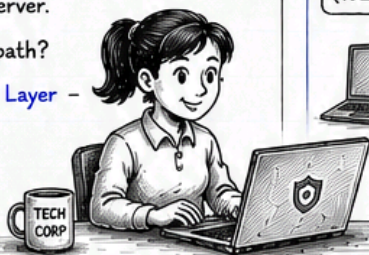
★ THE STORY SO FAR...

Transport Layer broke the message into segments and chose the best way to deliver them.

Now the message leaves Neha's computer and must travel across networks to reach the Bank Server.

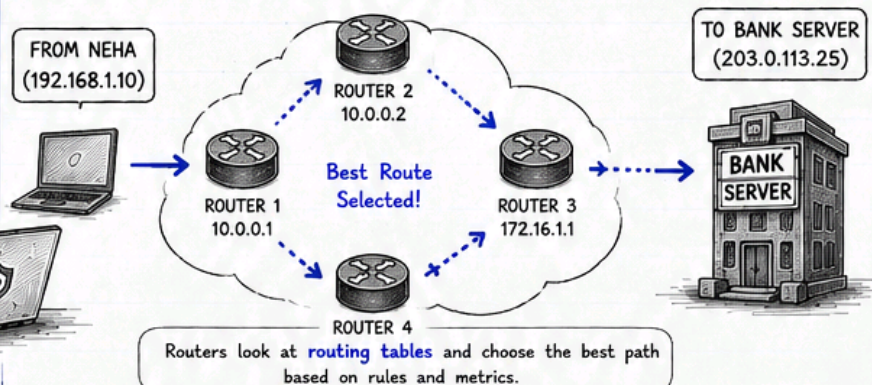
Who decides the path?

Meet the **Network Layer** – the Navigator!



IN THE MOVIE...

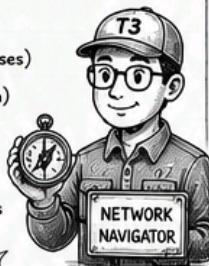
Neha's message (now in segments) leaves her computer. The Network Layer adds logical addresses (IP) and finds the best route to the bank server using routers.



★ WHAT IS NETWORK LAYER?

The Network Layer is **Layer 3** of the OSI Model. It is responsible for:

- ✓ Logical Addressing (IP Addresses)
- ✓ Routing (Finding the best path)
- ✓ Packet Forwarding
- ✓ Fragmentation (if needed)
- ✓ Handling congestion and errors



"I read the addresses and choose the best path for the packets!"

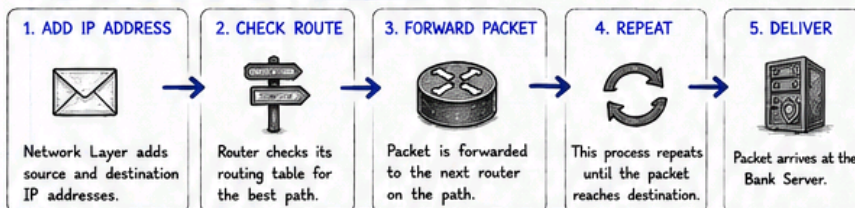
★ KEY CONCEPTS

- IP** **IP ADDRESSING**
Identifies devices across networks (IPv4 / IPv6).
- ROUTING**
Finds the best path to the destination network.
- ROUTING TABLE**
A table that stores paths for different networks.
- FRAGMENTATION**
Splits packets if they are too large for a network.
- TTL (TIME TO LIVE)**
Prevents packets from circulating forever.

★ EXAMPLES OF NETWORK LAYER PROTOCOLS

PROTOCOL	DESCRIPTION
IP (v4/v6)	Provides logical addressing and routing.
ICMP	Used for error reporting and diagnostics (e.g., ping).
IGMP	Manages multicast group memberships.
OSPF	A routing protocol used inside networks.
BGP	Routes data between networks (Internet).

★ HOW IT WORKS (IN SIMPLE STEPS)



IN THE MOVIE...

DarkGhost tries to send the message to a fake location (wrong IP). But the Network Layer uses routing logic to ignore wrong paths and send it on the right route.

"I can try to mislead it, but smart routing will find the real destination!"



★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts monitor:

- ✓ Unusual routing changes
- ✓ Suspicious IP addresses
- ✓ Data exfiltration attempts
- ✓ Routing attacks (e.g., ICMP redirect, BGP hijack)
- ✓ Abnormal traffic patterns

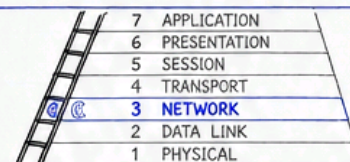
COMMON ATTACKS AT THIS LAYER

- IP Spoofing**
Attacker fakes the source IP address.
- ICMP Redirect Attack**
Tricks the system to take a longer or malicious route.
- BGP Hijacking**
Redirects traffic to an attacker's network.



KEY TAKEAWAY

The Network Layer is the Navigator. It uses IP addresses and routing to find the best path across networks and deliver the message to the right destination.



It finds the route.
It moves the packet.
It keeps the journey on track!





SCENE 7:

DATA LINK LAYER

"THE LOCAL TRAFFIC OFFICER"



★ THE STORY SO FAR...

The message has been routed across networks and has arrived at the local network of the Bank.

Now it needs to reach the correct device inside this network.

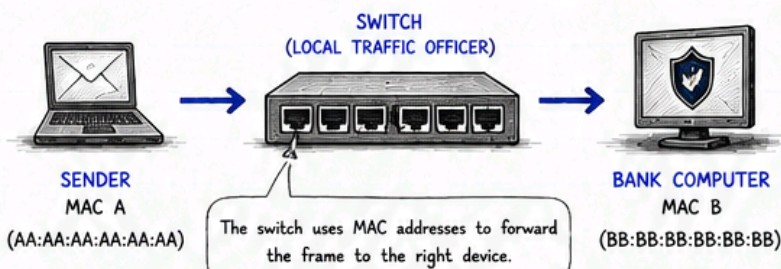
The Data Link Layer handles this local delivery using MAC addresses and frames.

Meet the **Data Link Layer** - the Local Traffic Officer!



IN THE MOVIE...

The message has reached the Bank's local network. The Data Link Layer looks at the MAC address and makes sure the frame is delivered to the correct device.



★ WHAT IS DATA LINK LAYER?

The Data Link Layer is Layer 2 of the OSI Model. It is responsible for:

- ✓ Node-to-Node delivery
- ✓ Framing the data
- ✓ MAC (Physical) Addressing
- ✓ Error detection and correction
- ✓ Flow control on the local network



Think of it as the local traffic officer who ensures safe delivery on the local road!

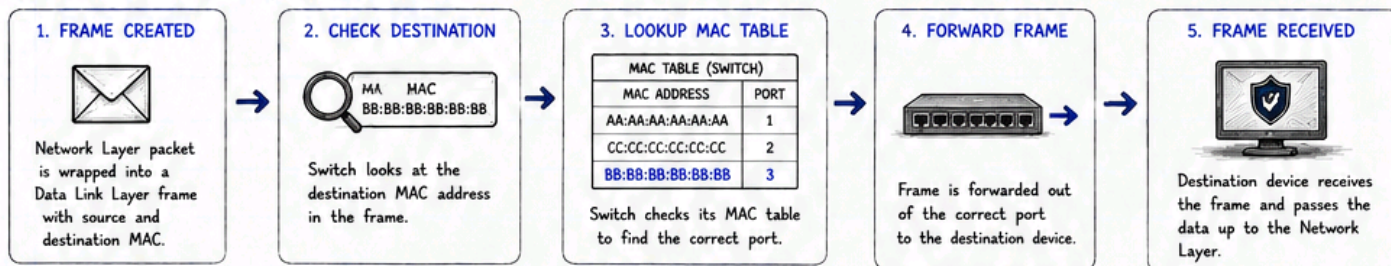
★ KEY CONCEPTS

MAC ADDRESS	Unique hardware address assigned to each device on a network.
FRAMES	Data is wrapped into frames for transmission.
SWITCHES	Forward frames based on MAC addresses.
ERROR CHECK	Detects and corrects errors using FCS/CRC.
ARP	Resolves IP address to MAC address.

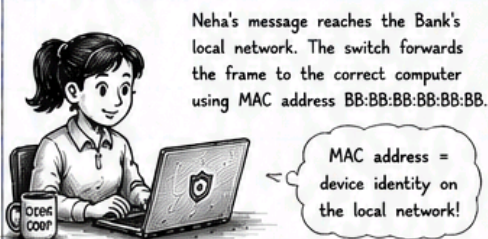
COMMON PROTOCOLS AT THIS LAYER

PROTOCOL	DESCRIPTION
Ethernet (IEEE 802.3)	Most common LAN technology that operates at Data Link Layer.
PPP	Used for point-to-point connections.
ARP	Maps IP address to MAC address.
STP	Spanning Tree Protocol - prevents network loops.

★ HOW IT WORKS (IN SIMPLE STEPS)



IN THE MOVIE...



HACKER SIENE...

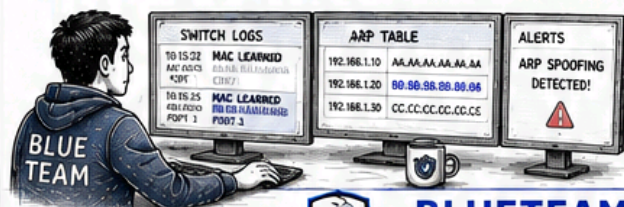
DarkGhost tries ARP Spoofing to trick devices and intercept frames.



COMMON ATTACKS AT THIS LAYER

- ⚠ **ARP Spoofing**
Tricks devices by sending fake ARP replies.
- ⚠ **MAC Flooding**
Overwhelms the switch MAC table.
- ⚠ **MAC Spoofing**
Attacker changes their MAC address to impersonate another device.
- ⚠ **DHCP Starvation**
Exhausts IP addresses to block access.

★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts monitor:

- ✓ ARP tables
- ✓ MAC address changes
- ✓ Switch logs
- ✓ Unusual device activity
- ✓ Traffic anomalies



KEY TAKEAWAY

The Data Link Layer is the Local Traffic Officer. It ensures frames are delivered correctly on the local network using MAC addresses.

It's all about local delivery, identity, and integrity!



SCENE 8:

PHYSICAL LAYER

"THE ROAD BENEATH EVERYTHING"

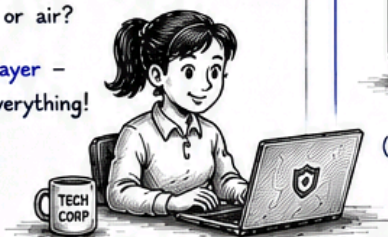


★ THE STORY SO FAR...

The message has traveled through many layers and has reached the Data Link Layer. Now it's ready to be sent as raw bits over the physical medium to the destination device.

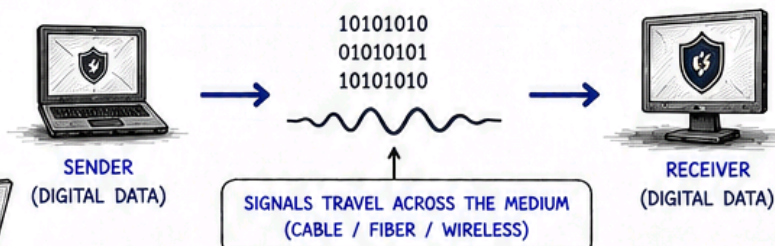
Who actually moves those bits as signals across cables, fiber, or air?

Meet the **Physical Layer** – the Road Beneath Everything!



IN THE MOVIE...

The Data Link Layer frame is turned into electrical, optical, or wireless signals (bits). These signals travel through the physical medium (cable, fiber, or air) and reach the destination device.



★ WHAT IS PHYSICAL LAYER?

The **Physical Layer** is **Layer 1** of the OSI Model. It is responsible for moving raw bits from one device to another over the physical medium.

It deals with:

- ✓ Bit transmission
- ✓ Electrical / Optical / Radio signals
- ✓ Hardware specifications
- ✓ Cables, connectors, and voltages
- ✓ Data rate and synchronization



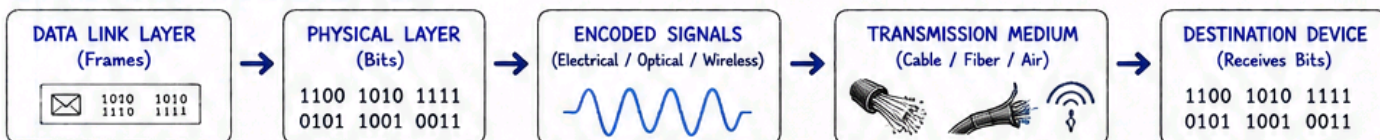
★ KEY CONCEPTS

0101 1010	BITS	Data in 0s and 1s.
	SIGNALS	Electrical / Optical / Radio waves.
	MEDIUM	Copper Cable, Fiber Optic, or Wireless.
	DEVICES	Hubs, Repeaters, Cables, Antennas.
1010 1011	ENCODING	Converts bits to signals (e.g., NRZ).
	SYNCHRONIZATION	Keeps sender and receiver in sync.

TRANSMISSION MEDIUMS

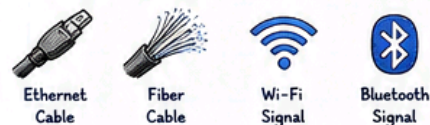
	COPPER CABLE (Twisted Pair)	Uses electrical signals. Common in LANs and telephones.
	FIBER OPTIC CABLE	Uses light signals. Fast, reliable, and immune to interference.
	WIRELESS (RADIO)	Uses radio waves. No physical cable needed.

IN THE MOVIE: THE SIGNALS FLOW



IN THE REAL WORLD...

Your devices rely on the Physical Layer every second to communicate.



HACKER SCENE...

DarkGhost can tap cables, jam signals, or use rogue devices to intercept or disrupt communication at this layer.

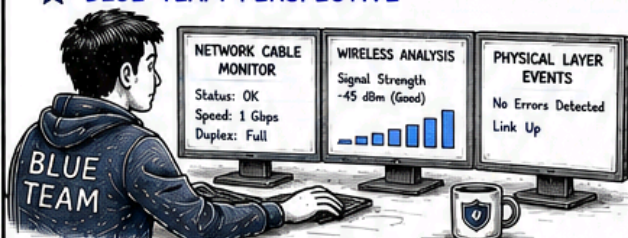


COMMON ATTACKS AT THIS LAYER

- ⚠ Cable Tapping – Physically intercepting signals.
- ⚠ Jamming – Blocking wireless signals.
- ⚠ Rogue Devices – Using unauthorized hardware (e.g., fake AP).
- ⚠ Signal Interference – Introducing noise to corrupt signals.



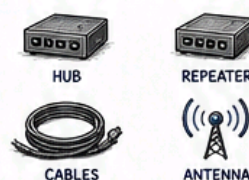
★ BLUE TEAM PERSPECTIVE



At this layer, Blue Team analysts monitor:

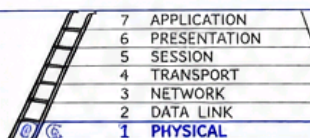
- ✓ Link status and errors
- ✓ Signal strength and quality
- ✓ Cable integrity
- ✓ Rogue device detection
- ✓ Physical access to devices

DEVICES AT THIS LAYER



KEY TAKEAWAY

The Physical Layer is the foundation of all communication. It moves raw bits as signals through cables, fiber, or air. Without the road, the journey cannot begin!



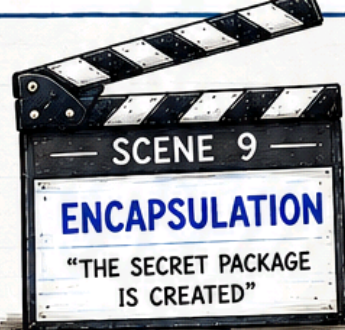
It's the road beneath everything – invisible but absolutely essential!



SCENE 9:

ENCAPSULATION

"THE SECRET PACKAGE IS CREATED"



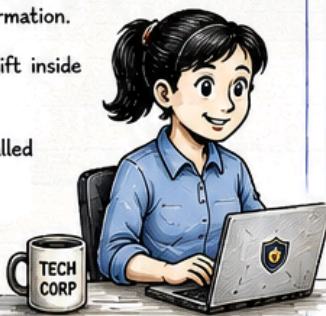
THE STORY SO FAR...

Neha's message has passed through all the OSI layers from Application down to Physical Layer.

Now, before the message travels across the network, every layer adds its own information.

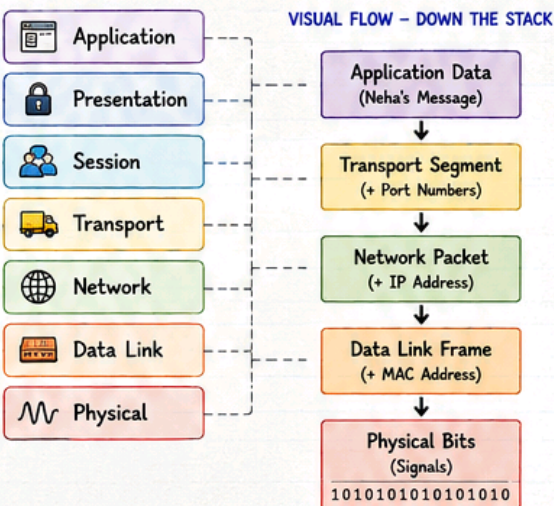
Like wrapping a gift inside multiple boxes!

This process is called **Encapsulation**.

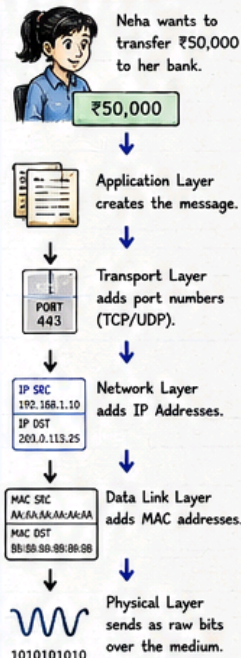


WHAT IS ENCAPSULATION?

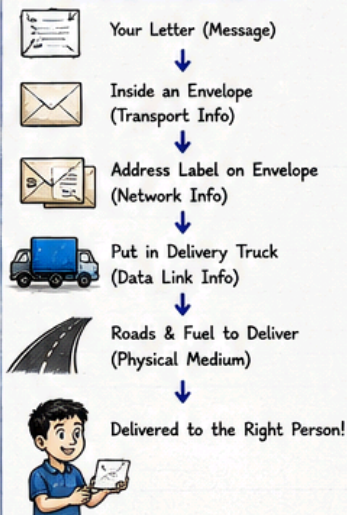
Encapsulation is the process of adding headers (and sometimes trailers) while moving down the OSI layers from Layer 7 to Layer 1. Each layer adds its own information to help the data reach the destination.



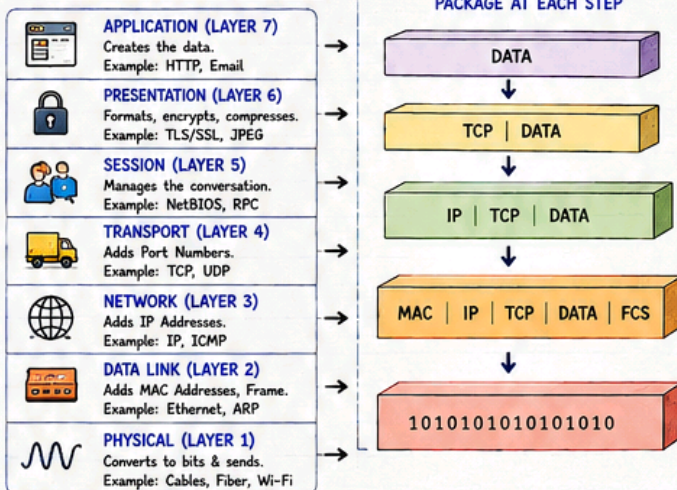
MOVIE SCENE



5TH CLASS EXAMPLE

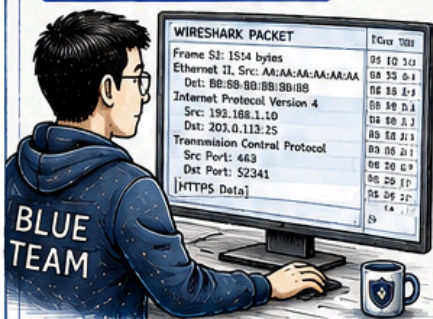


ENCAPSULATION IN ACTION



- Only Data (Application)
- - - + Port Numbers (Transport Header)
- - - + IP Addresses (Network Header)
- - - + MAC Addresses & Trailer (Frame)
- - - Converted to Bits (Signals)

BLUE TEAM PERSPECTIVE



Encapsulation helps analysts understand the structure of the packet.

We can see each layer's information in the packet capture tool like Wireshark!

WHY IS ENCAPSULATION IMPORTANT?

- ✓ Ensures data is delivered correctly at each layer.
- ✓ Helps in routing, delivery, and error handling.
- ✓ Allows communication between different devices and networks.
- ✓ Provides structure to the raw data.

REAL WORLD EXAMPLE



KEY TAKEAWAY

Encapsulation is like preparing a package for a long journey. Each layer adds the right label, address, and protection so the message reaches the correct destination.

“Encapsulation prepares data for travel. Without it, the message would get lost!”





SCENE 10:

DECAPSULATION

"OPENING THE SECRET PACKAGE"

THE STORY SO FAR...

Neha's message has traveled all the way through the network and has reached the Bank Server.

Now the journey is in reverse.

Each OSI layer removes its own information so the original message can be read.

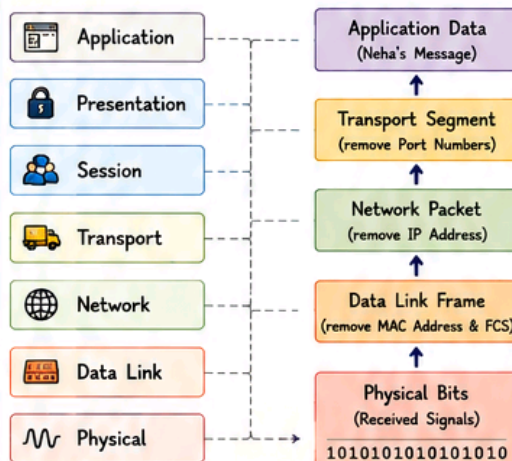
This process is called **Decapsulation**.



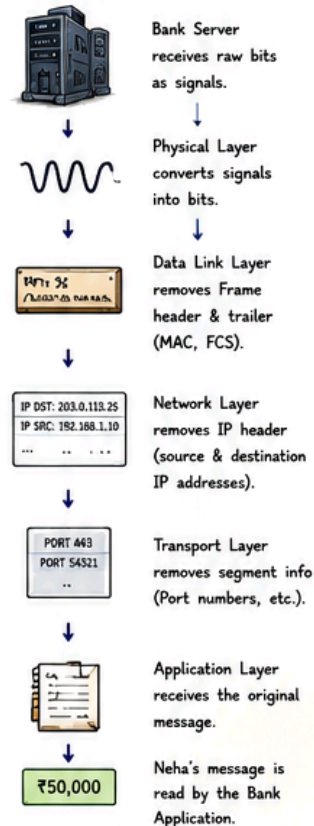
WHAT IS DECAPSULATION?

Decapsulation is the reverse process of Encapsulation. Each layer removes the header (and trailer) it added as the data moves up the OSI stack.

VISUAL FLOW - UP THE STACK (AT DESTINATION)



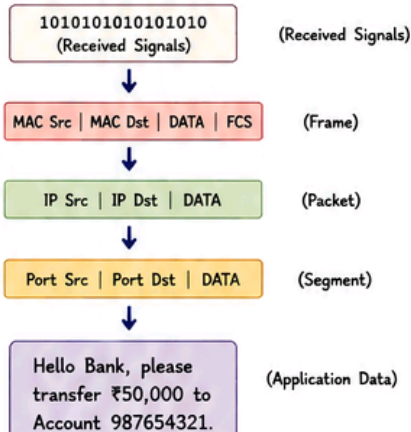
MOVIE SCENE - AT BANK



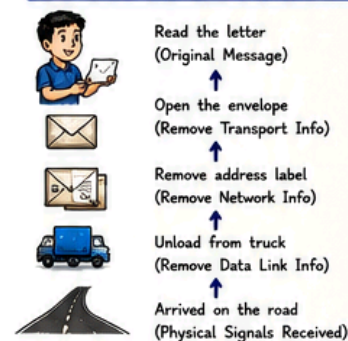
DECAPSULATION IN ACTION

LAYER	ACTION (AT DESTINATION)	REMOVES
Application	Receives data and presents it to the user.	-
Presentation	Decodes, decrypts, decompresses, data if needed.	Presentation Header
Session	Manages session and removes session info.	Session Header
Transport	Reassembles segments and removes port numbers.	Transport Header
Network	Removes IP header and finds the correct application.	Network Header
Data Link	Checks FCS, removes MAC header and trailer.	Data Link Header and Trailer (FCS)
Physical	Converts signals into bits and delivers to Data Link.	-

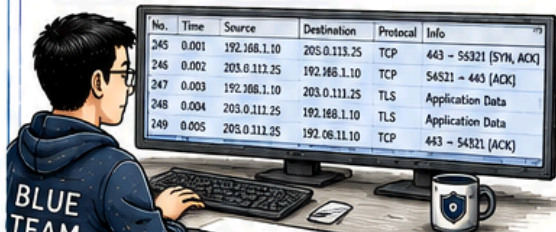
FROM BITS TO MESSAGE



STH CLASS EXAMPLE - IN REVERSE



BLUE TEAM PERSPECTIVE



Decapsulation happens inside the system and tools like Wireshark show us each layer's information.

We can analyze packets layer by layer to understand what really happened!

WHY IS DECAPSULATION IMPORTANT?

- ✓ Delivers the original message to the correct application.
- ✓ Ensures data is understood and processed correctly.
- ✓ Helps in troubleshooting and root cause analysis.
- ✓ Essential for security monitoring and threat detection.

REAL WORLD EXAMPLE



After decapsulation, the Bank Application reads Neha's message and the money is transferred successfully!

MESSAGE DELIVERED SUCCESSFULLY!



KEY TAKEAWAY

Decapsulation is like unwrapping a gift. Each layer removes what it added, so the original message can be used.

“Decapsulation delivers the message safely from the network to the application.”



SCENE 11:

THE HACKER ATTACKS

"DARKGHOST MAKES HIS MOVE"

SCENE 11

"THE HACKER ATTACKS
"DARKGHOST"
MAKES HIS MOVE"

THE STORY SO FAR...

Neha is transferring ₹50,000 to her bank.

DarkGhost, a cyber criminal, wants to intercept, steal, or manipulate this transaction.

He knows that every communication passes through **7 OSI layers**.

So, he tries to **attack** at **every possible layer**!

I'll attack each layer one by one!

ATTACKS ACROSS THE OSI MODEL

7	APPLICATION LAYER		Phishing / Malicious Websites Tricks users, steals credentials, injects malware.
6	PRESENTATION LAYER		Weak Encryption / Data Exposure Exploits weak encryption, steals data in transit.
5	SESSION LAYER		Session Hijacking Steals active sessions, takes over connections.
4	TRANSPORT LAYER		Port Scanning / TCP Flood Finds open ports, overwhelms services.
3	NETWORK LAYER		IP Spoofing / Route Hijack Fakes IP addresses, redirects traffic.
2	DATA LINK LAYER		ARP Spoofing / MAC Spoofing Poisons ARP cache, intercepts/redirects frames.
1	PHYSICAL LAYER		Cable Tapping / Jamming Sniffs raw signals, jams or disrupts connections.

! DarkGhost can attack at any layer to steal, alter, or disrupt the message.

DARKGHOST'S ATTACK PLAYBOOK

	PHISHING Sends fake emails or creates fake websites to steal credentials.
	WEAK ENCRYPTION Forces weak ciphers or exploits unencrypted data.
	SESSION HIJACKING Steals session cookies or tokens to take over a user's session.
	PORT SCANNING & TCP FLOOD Scans for open ports and floods services to cause outage.
	IP SPOOFING Fakes IP address to pretend to be a trusted system.
	ARP SPOOFING Sends fake ARP replies to intercept or redirect traffic.
	CABLE TAPPING & JAMMING Sniffs network cables or jams wireless signals.

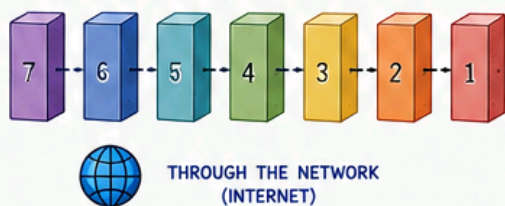
DARKGHOST VS THE MESSAGE

NEHA (SENDER)

MESSAGE TRAVELS THROUGH OSI LAYERS

DARKGHOST (ATTACKER)

BANK SERVER (RECEIVER)

₹50,000
MESSAGETRIES TO INTERCEPT,
STEAL OR MANIPULATE₹50,000
MESSAGE

WHY ATTACK MULTIPLE LAYERS?

- ✓ If one attack fails, try another layer.
- ✓ Different attacks target different weaknesses.
- ✓ Sometimes, lower layer attacks are silent and hard to detect.
- ✓ Defense must be strong at every layer!

REAL WORLD EXAMPLES

- ✉ Phishing Email → Stolen Bank Credentials
- 🔗 ARP Spoofing → Man-in-the-Middle Attack
- 👤 IP Spoofing → Bypass Access Controls
- 💣 TCP Flood → Website Down (DoS)
- 🔍 Cable Tapping → Data Leakage

KEY TAKEAWAY

Attackers like DarkGhost can target any OSI layer.
Strong security means protecting every layer, not just the top!

"You can't defend what you don't understand."
– Blue Team Motto

BLUE TEAM PERSPECTIVE



Understanding attacks at each OSI layer helps Blue Team analysts:

- ✓ Detect threats faster
- ✓ Investigate deeper
- ✓ Block attacks effectively
- ✓ Build stronger defenses

REMEMBER

The OSI Model is not just theory. It is your map to understand, detect, and stop attacks at every level!

KNOW THE LAYERS. STOP THE ATTACKS.



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DEFEND. DETECT. RESPOND.



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SCENE 12:

BLUE TEAM WAR ROOM

"ALEX INVESTIGATES"



THE STORY SO FAR...

DarkGhost has launched his attack.
But the Blue Team is ready.

ALERT!

Suspicious traffic
detected!

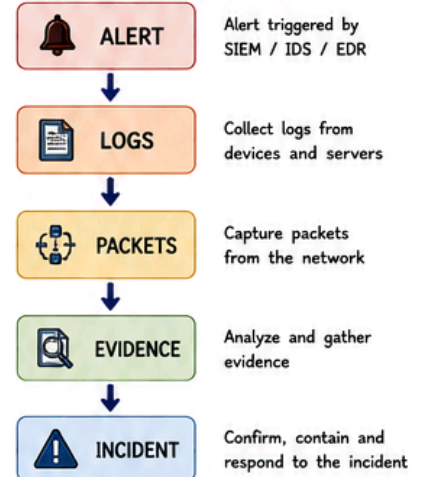
Alex, our Blue Team
analyst, jumps into
the War Room and
starts the investigation.



TOOLS ALEX USES

	Wireshark	Captures and analyzes network packets.
	Splunk	Searches and analyzes logs and events.
	Microsoft Sentinel	SIEM / SOAR for threat detection and response.
	EDR / XDR	Detects and responds to threats on endpoints.

INVESTIGATION FLOW



WHAT ALEX CHECKS

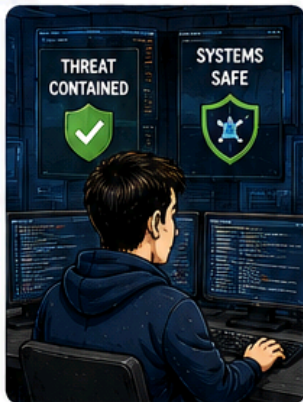
WHAT	DESCRIPTION	EXAMPLE
Source IP	Where the traffic is coming from.	203.0.113.25
Destination IP	Where the traffic is going.	192.168.1.10
Protocol	Type of communication (TCP, UDP, ICMP, etc.)	TCP
Port	Service or application being used.	443 (HTTPS)
MAC Address	Device hardware address at Layer 2.	AA:BB:CC:DD:EE:FF
Time / Timestamp	When the activity happened.	2025-05-16 14:25:33
Geolocation	Where the IP is located.	Singapore
Log Events	Related events from devices.	Failed Login, Suspicious Process

ALEX IN ACTION - WAR ROOM



BLUE TEAM PERSPECTIVE

- ✓ We don't just see alerts. We investigate them.
- ✓ We connect the dots across all layers.
- ✓ We find the attacker's path and stop the damage.
- ✓ We protect the users, the data and the business.



REAL WORLD EXAMPLE



TOOLS IN ACTION

	Wireshark	• Capture Live Traffic • Analyze Packets
	Splunk	• Search Logs • Create Dashboards
	Microsoft Sentinel	• Correlate Alerts • Automated Response
	EDR / XDR	• Endpoint Visibility • Threat Hunting



KEY TAKEAWAY

The OSI Model gives context.
The tools give visibility.
The Analyst connects the dots.

“

OSI helps Blue Team analysts
investigate attacks faster and respond better.”

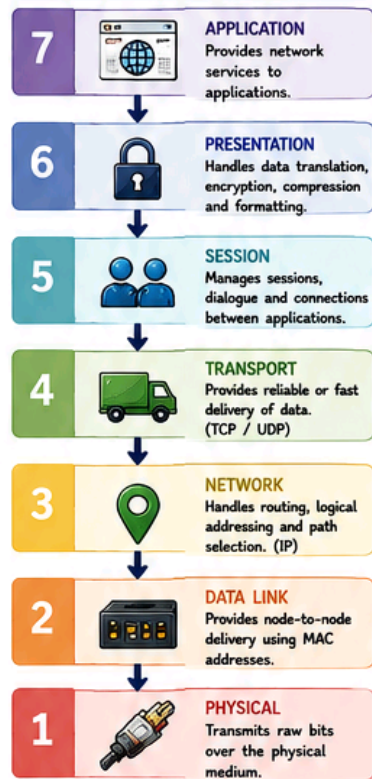


One Page.
One Review.
One Step Closer to
Becoming a
Blue Team Analyst!"

OSI MODEL MASTER CHEAT SHEET

★ EVERYTHING ON ONE PAGE ★

THE 7 LAYERS OF OSI MODEL



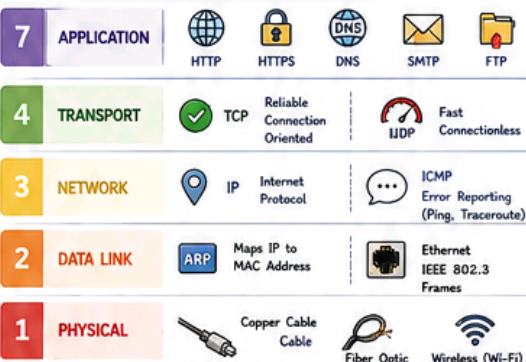
LAYER SUMMARY TABLE

LAYER	NAME	JOB (MAIN FUNCTION)	PDU	KEY INFO
7	Application	Provides user services.	Data	Examples: HTTP, HTTPS, DNS, FTP, SMTP
6	Presentation	Translates, encrypts and formats data.	Data	Handles encryption, compression, encoding.
5	Session	Establishes, manages and terminates sessions.	Data	Manages dialogue between systems.
4	Transport	Ensures delivery end-to-end.	Segment	TCP (Reliable) UDP (Fast)
3	Network	Routes data across networks.	Packet	Uses IP Address for routing.
2	Data Link	Delivers data to next node.	Frame	Uses MAC Address for local delivery.
1	Physical	Transmits raw bits as signals.	Bits	Electrical, Optical or Wireless signals.

DEVICES USED IN OSI



PROTOCOL CHEAT SHEET

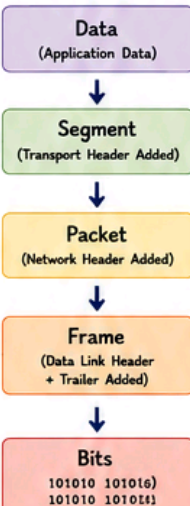


BLUE TEAM QUICK NOTES



Great Analysts Understand Every Layer.
Great Defenders Protect Every Layer.

ENCAPSULATION FLOW (SENDING SIDE)



DECAPSULATION FLOW (RECEIVING SIDE)



MEMORY TRICK

All
People
Seem
To
Need
Data
Processing



INTERVIEW QUESTIONS

- What is OSI Model? ?
- Which layer uses IP Address? ?
- Which layer uses MAC Address? ?
- What is the difference between TCP and UDP? TCP + UDP
- What is Encapsulation? ?
- What is Decapsulation? ?



KEY TAKEAWAY

OSI Model is the foundation of Networking and Cyber Security. Understand each layer and you will understand how the internet works and how to defend it!



REVISE. PRACTICE. PROTECT.

Master OSI Today,
Become a Blue Team Hero Tomorrow! ❤️



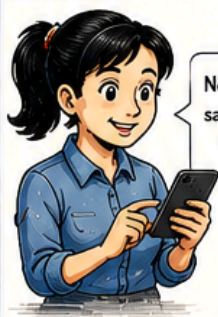
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DEFEND. DETECT. RESPOND.



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THE END

THE COMPLETE JOURNEY OF A MESSAGE



Neha's transaction safely reaches the Bank Server.

✓ **MISSION SUCCESSFUL!**

SENDING SIDE (NEHA'S DEVICE)

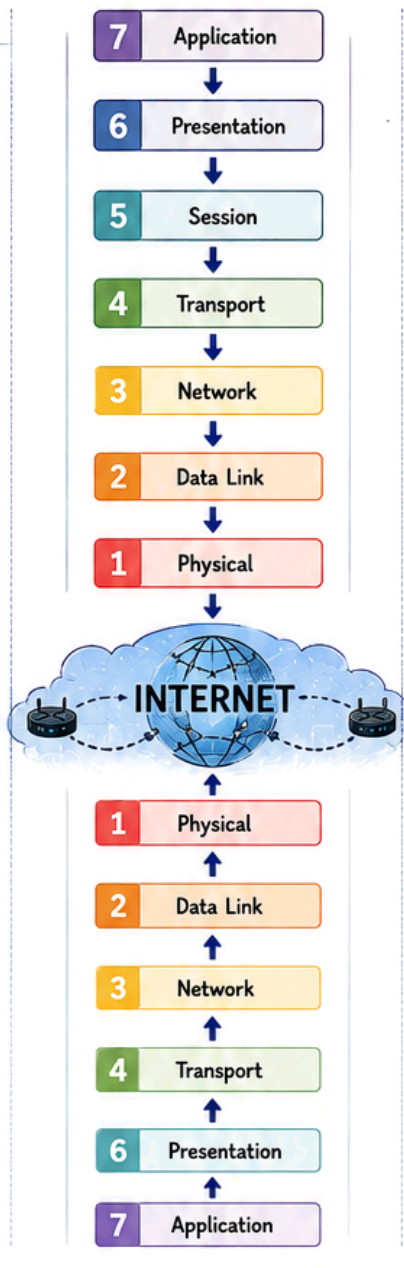


LESSONS LEARNED

- ✓ What OSI Model Is
- ✓ 7 Layers of OSI
- ✓ TCP vs UDP
- ✓ Routing using IP
- ✓ MAC Address & Local Delivery
- ✓ Signals & Transmission
- ✓ Encapsulation
- ✓ Decapsulation
- ✓ Cyber Attacks Across Layers
- ✓ Blue Team Investigation & Tools

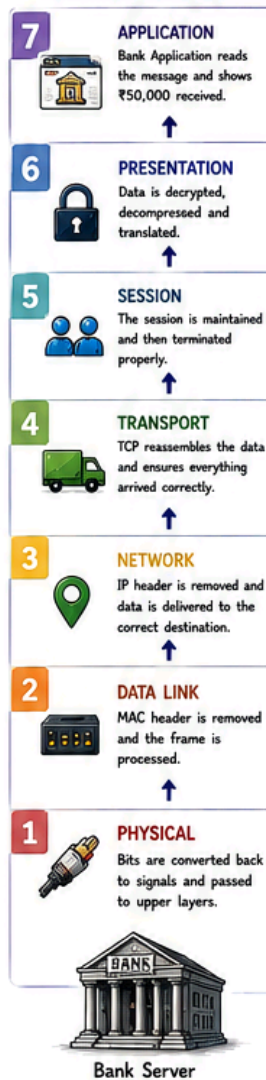


FULL OSI JOURNEY (END-TO-END)



DarkGhost fails.
Alex and the Blue Team win!
The organization is protected.

RECEIVING SIDE (BANK SERVER)



BLUE TEAM WINS!

CAREER CONNECTIONS

- SOC Analyst**
Monitor, Detect and Respond
- Network Analyst**
Understand Networks and Protocols
- Security Analyst**
Analyze Threats and Vulnerabilities
- Incident Responder**
Investigate Incidents and Contain Threats
- Threat Hunter**
Proactively Hunt for Threats
- Cloud Security Analyst**
Secure Cloud Infrastructure

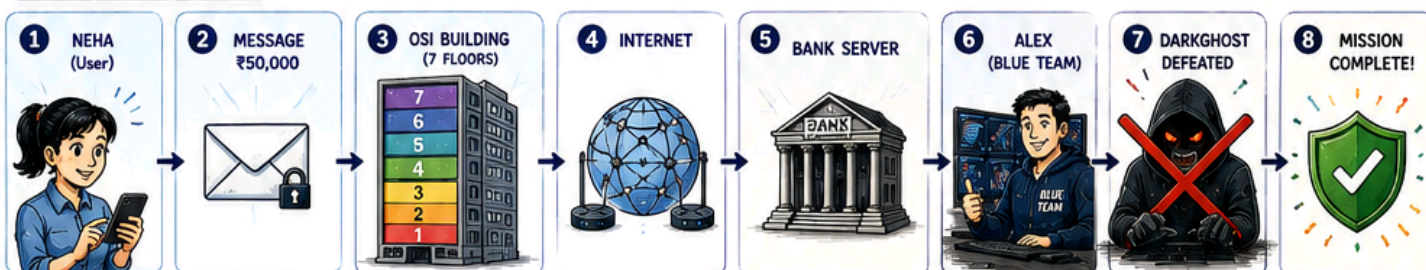
The OSI Model is the foundation for many Cyber Security careers!



“Every packet has a story.

Understanding the OSI Model helps Blue Team defenders understand where the packet came from, where it is going, and how to stop attackers along the way.”

THE MOVIE ENDING SCENE



FINAL TAKEAWAY

OSI Model turns complex networking into simple, logical steps.

LEARN IT. PRACTICE IT. DEFEND IT.
BE THE BLUE TEAM HERO! ♥



BLUETEAMERS
DEFEND. DETECT. RESPOND.